

modified FoldX  $\Delta\Delta G$  (kcal/mol)

25  
20  
15  
10  
5  
0

IA138

SA139

RA140

CA141

FA142

TA143

RA144

KA145

YA146

TA147

LA148

PA149

PA150

GA151

VA152

DA153

PA154

TA155

LA156

VA157

SA158

SA159

SA160

LA161

SA162

PDEGSSIZ

PDGTHSSIZ

P

X

GA

PDGSSD

G

X

X

X

X

X

GS

X

PGC

X

P

P

X

PV